

Assignment 7: Ion Implantation and Silicidation

FinFET transistors are a crucial component of modern integrated circuits, offering superior performance compared to traditional planar transistors. However, their fabrication process requires precise control over doping and contact resistance to ensure optimal device characteristics. This report explores two key aspects of FinFET fabrication: the significance of wafer tilt and annealing temperature during ion implantation and the necessity of silicidation.

Question 1: Ion implantation

This study focuses on two key aspects: activation anneal temperature and wafer tilt angle during ion implantation. We determine the maximum activation anneal temperature to activate dopants while maintaining a 5nm channel length. Additionally, we explore the allowable tilt angle to ensure proper doping and investigate the effect of tilt on dopant distribution through concentration plots.

Ion implantation is a precise doping technique which involves the generation of ion. These ions are then electrostatically accelerated to high energies and directed towards a target material. Upon impact, the ions penetrate the target lattice, altering its properties.

Figure 1 shows process of ion implantation.

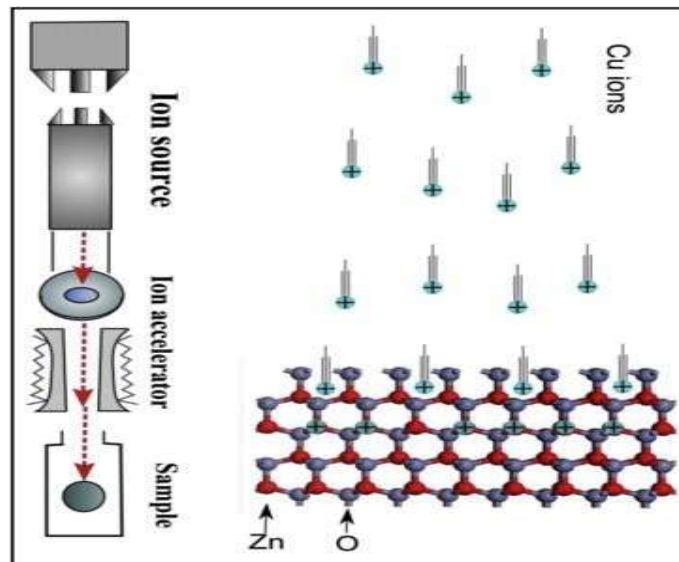


Figure 1 : Ion Implantation Process

Ion implantation and Silicidation done by process flow provided to us is shown by Figure 2 and Figure 3 shows the final structure.

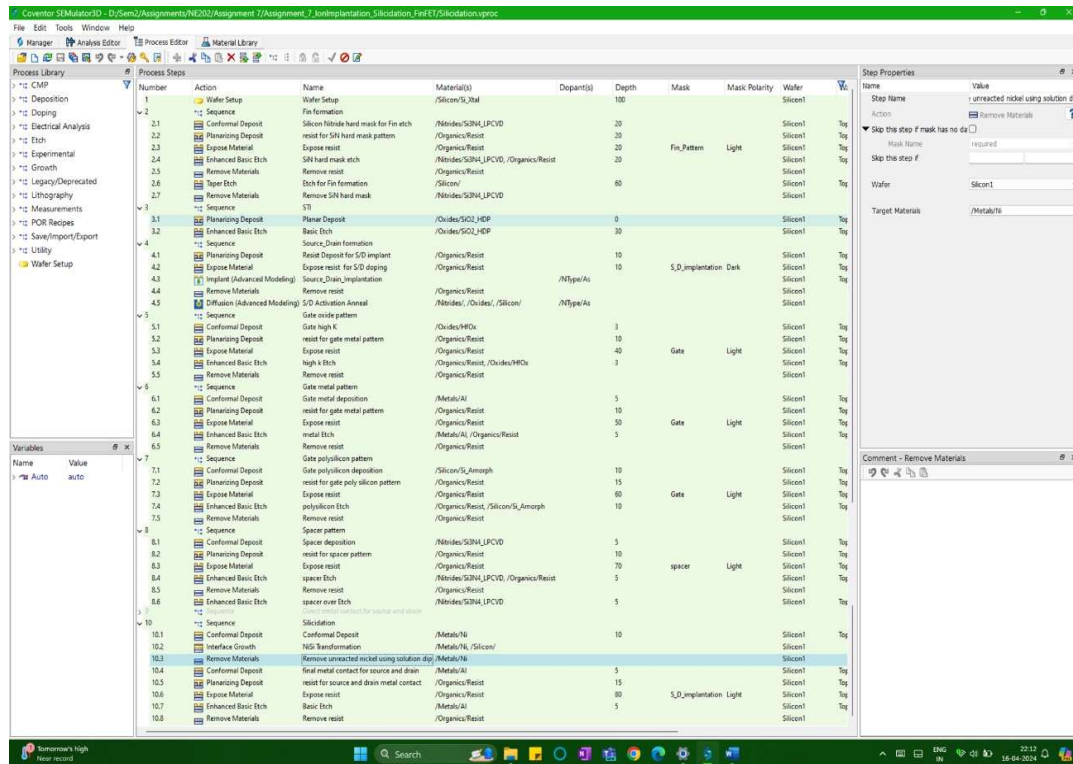


Figure 2 : Process flow for ion implantation & silicidation

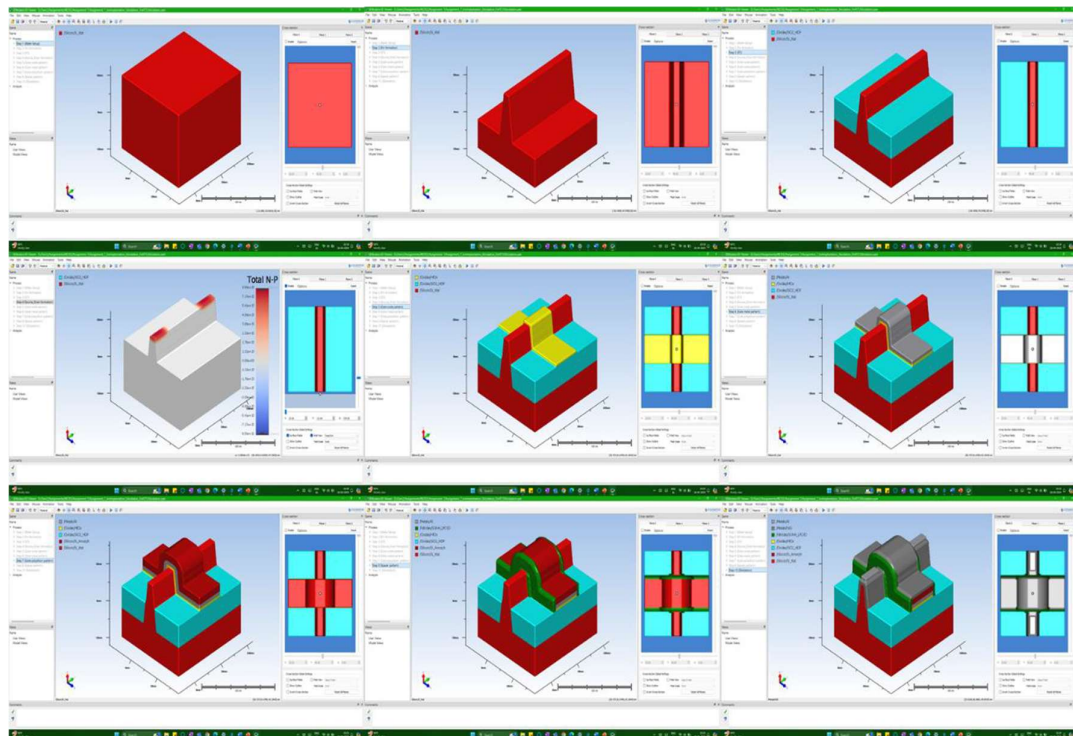


Figure 3 : Final structure after ion implantation & silicidation

For initial process Si substrate is intrinsic, which are unsuitable for n-channel FinFETs due to the lack of majority carriers makes it difficult to form an inversion layer and requires a very high gate voltage to turn on. Also, it offers nominal resistance to leakage current, leading to higher off-state current flow. If we choose N-type substrate which already have electrons as majority charge carriers will resist the formation of this channel and requires a higher gate voltage to turn on, whereas in p-type substrate lack of electrons allows the gate to efficiently create the channel and operate at a lower gate voltage. So for proper channel formation in N-channel FinFET, the substrate should be P-type.

Here the substrate is changed to P-type by doping initial wafer with Boron concentration of $1\text{e}13\text{ cm}^{-3}$ in step 1: wafer setup (Figure 4).

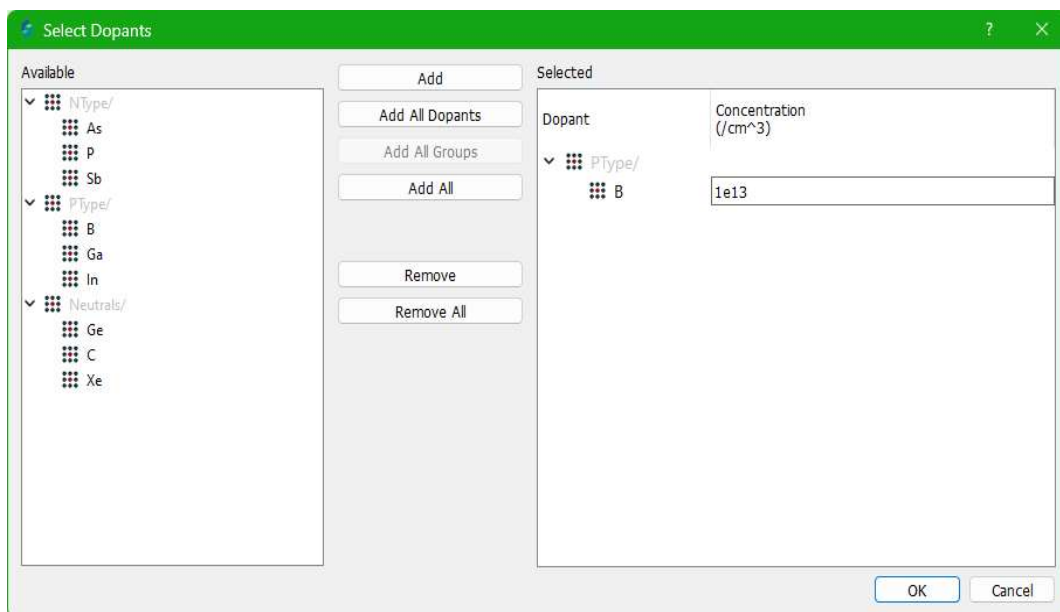


Figure 4 : Doping the initial Si wafer

Question1(a)

In this part we need to vary activation anneal temperature for source and drain and find optimal point for which channel length becomes 5 nm and diffuse beyond Fin height.

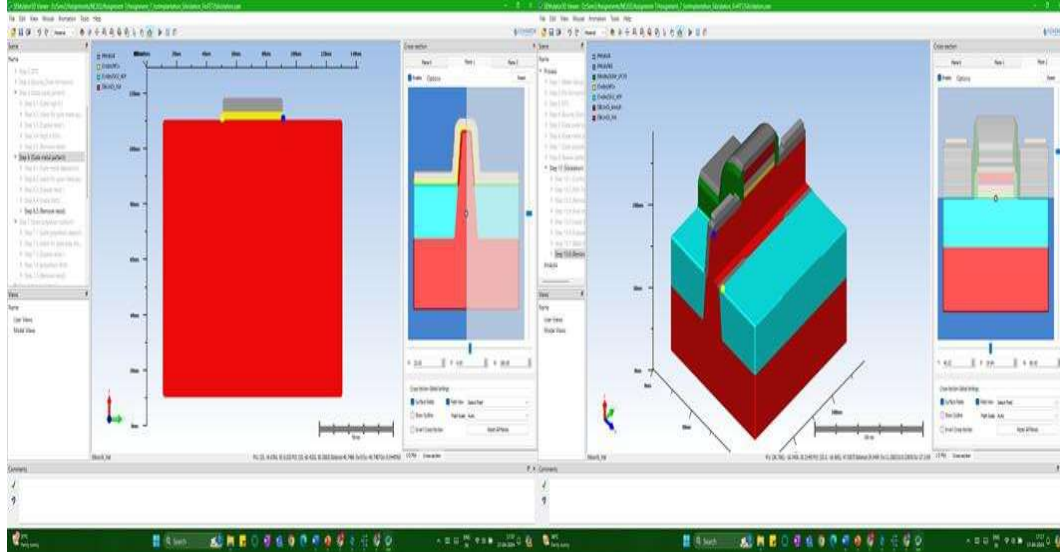


Figure 5 : Ideal channel width & Fin height

As shown in Figure 5, for measuring ideal channel length we measure gate oxide length (from $y = 20\text{nm}$ to $y = 60\text{nm}$) which comes out to be 40 nm and for Fin height we start measuring from top of cut fin until start of STI which is 30 nm .

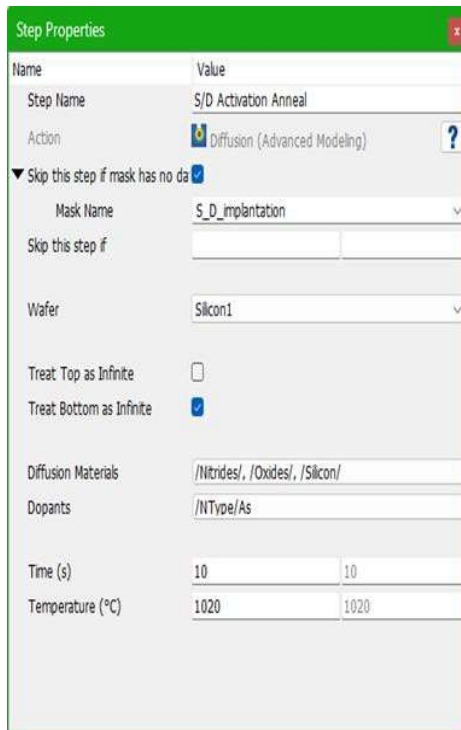


Figure 6 : Annealing properties

Table 1 : variation in annealing temperature

Sl.No.	Annealing Temperature (°C)	Channel length (nm)	Depleted Fin height (nm)
1.	1020	37.4	15.9
2.	1030	33.1	17.8
3.	1040	28.8	19.7
4.	1050	26.7	21.7
5.	1060	23.9	23.5
6.	1070	20.1	24.8
7.	1080	17.1	25.9
8.	1085	14.8	27.0
9.	1090	12.7	28.3
10.	1095	10.7	29.8
11.	1100	8.6	30.4
12.	1104	6.1	32.3
13.	1106	5.0	33.1
14.	1110	1.9	33.8

For varying annealing temperature, Temperature parameter is varied in step 4.5: Diffusion (Figure 6) in process flow according to Table 1.

Fin height is measured along z axis, it should be where the concentration of Total N-P becomes zero in fin depth below either source or drain.

For annealing temperature 1095°C whole fin height is consumed by dopants as seen in Figure 7. So, for preventing dopants to diffuse beyond fin height annealing temperature should be less than 1095°C.

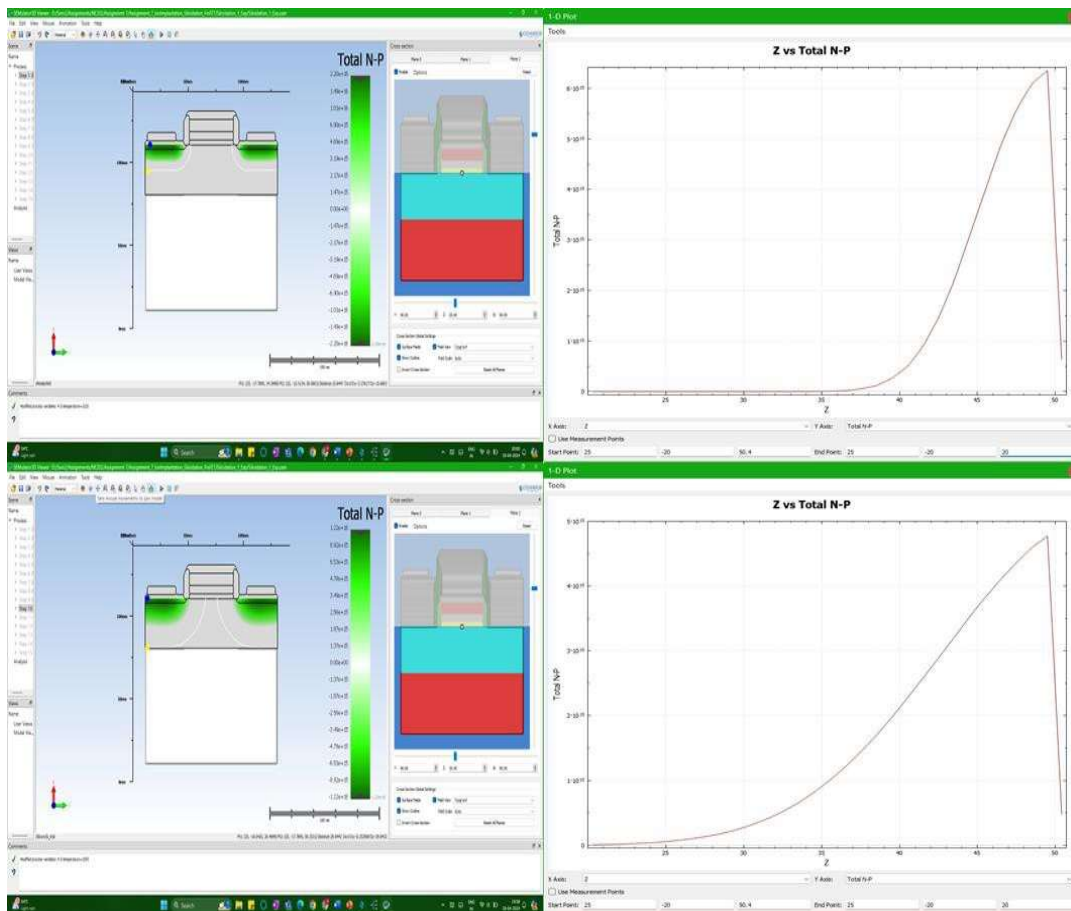


Figure 7 : Diffused dopant depth in fin

Channel length is measured along y axis, it will be the distance between source and drain depletion region below gate or wherever Total N-P becomes zero.

The temperature at which channel length becomes 5 nm is 1106°C (Figure 8), i.e. greater than temperature at which whole fin is depleted. So, to maintain channel length alone annealing temperature can go up to 1106°C.

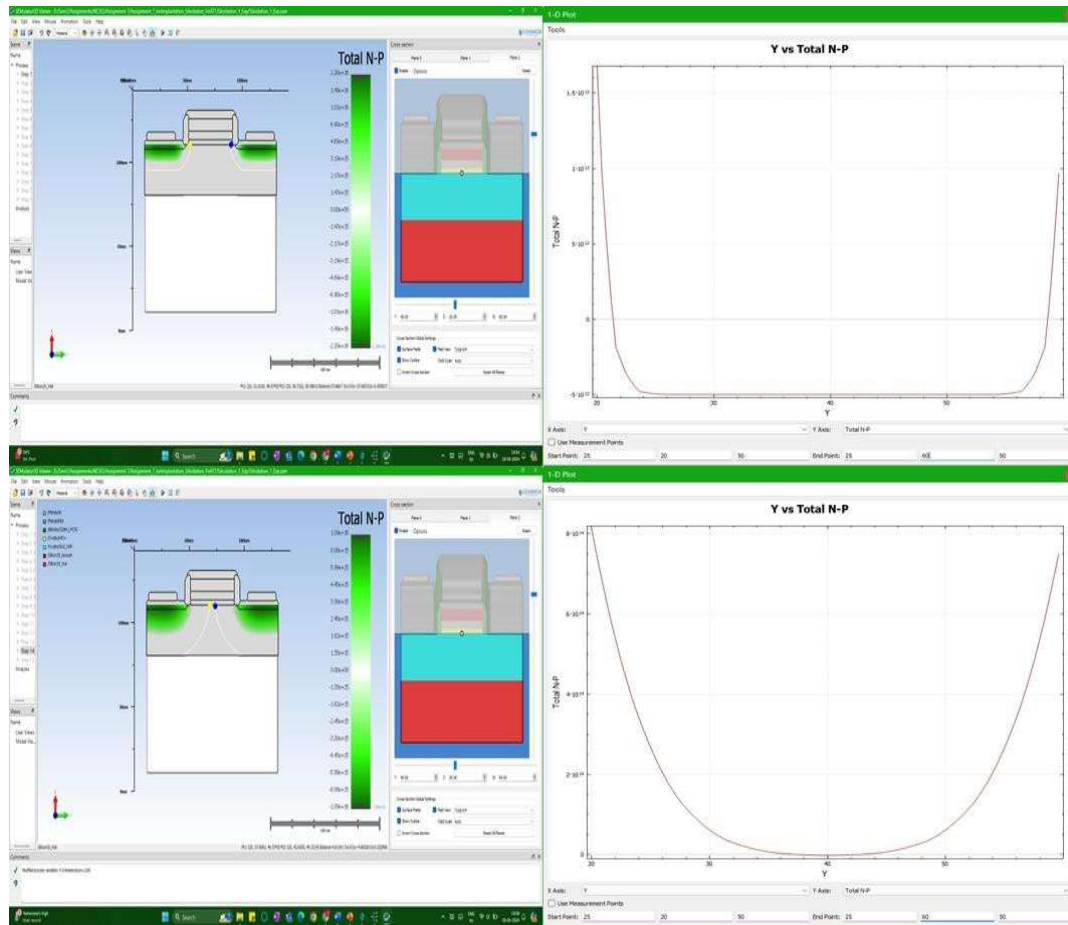


Figure 8 : Reduction in channel length

To maintain channel length 5 nm along with prevention of dopant diffusion beyond fin height, temperature for annealing should not exceed 1095°C.

Question1(b)

This question deals with the allowable tilt angle during ion implantation to ensure proper doping in the Fins. For this we vary tilt angle in step 4.3.

Table 2 : variation in tilt angle

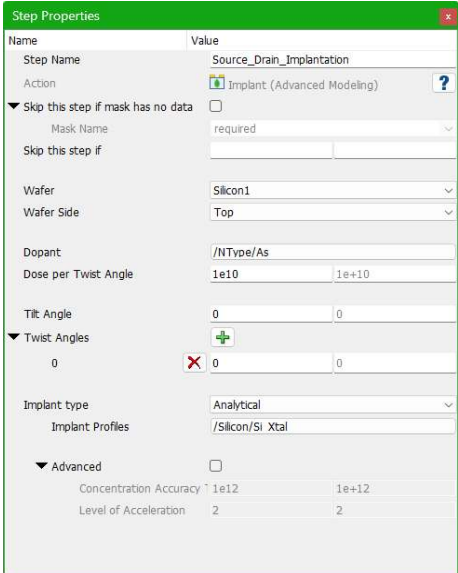


Figure 9 : Implantation properties

Sl.No.	Tilt angle
1.	0°
2.	1°
3.	2°
4.	3°
5.	4°
6.	5°
7.	6°
8.	7°

The doping profile is symmetric upto 5° tilt angle from around ~6°-7° it starts to get asymmetric.

So, the allowable tilt angle during ion implantation to ensure proper doping in the Fins is around ~6°-7°.

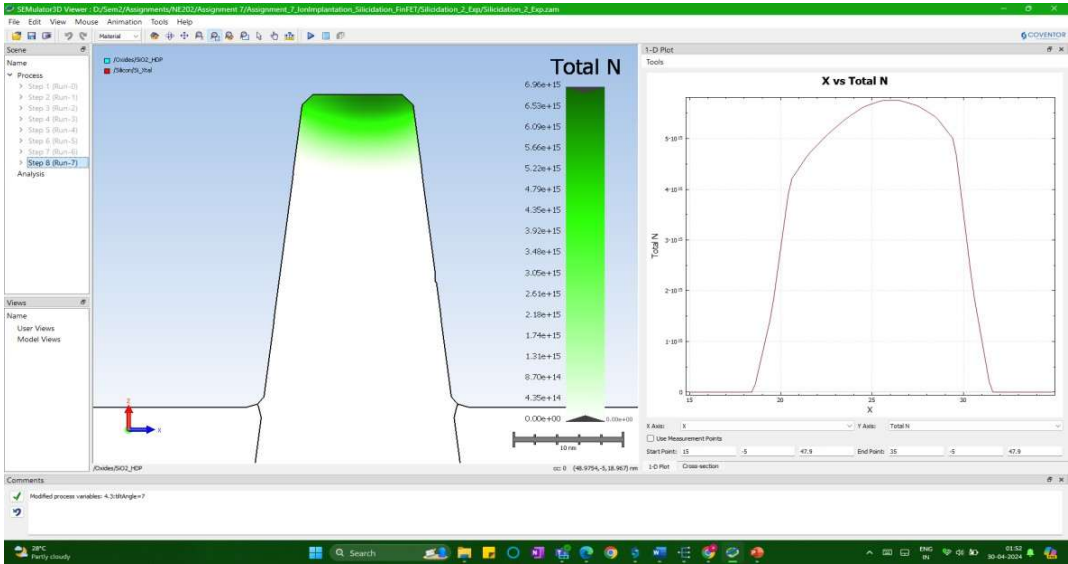


Figure 10 : Doping Profile for tilt angle = 7°

Question1(c)

Shadowing Effect:

When the ion beam is directed at the target material with a tilt angle, rather than a normal orientation, a phenomenon known as shadowing arises. This occurs because surface features like photoresist masks, gate electrodes, or fin structures act as partial barriers, intercepting a portion of the incoming ions.

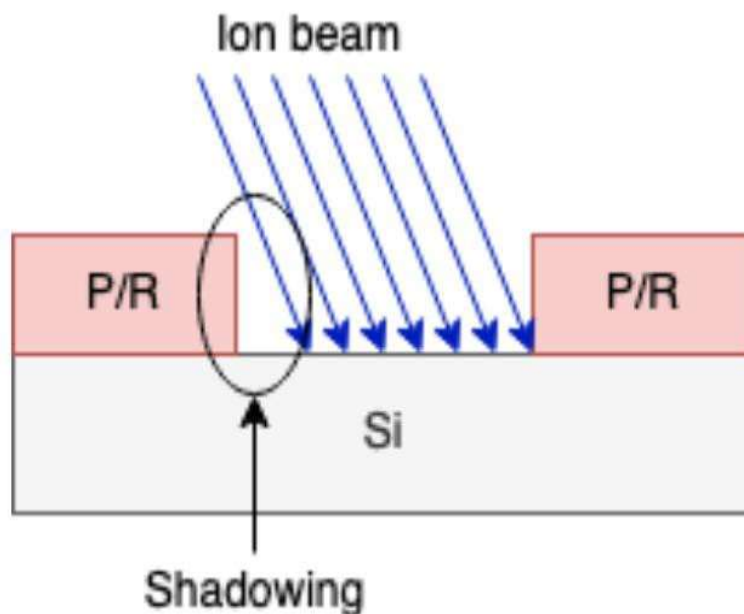


Figure 11 : Shadowing Effect

Consequently, these features create "shadowed" regions on the underlying material, resulting in a non-uniform distribution of implanted ions as shown by Figure 11. This translates to variations in dopant concentration across the material, with higher concentrations in non-shadowed areas and lower concentrations in shadowed regions.

Yes, with respect to centre of channel dopant distribution become asymmetrical due to tilt.

The tilt angle in SEMulator 3D tilts the ion source to given angle with respect to +x axis.

Dopant distribution across Fin width for a symmetric case (tilt angle = 0°) and an asymmetric case (tilt angle = 5°).

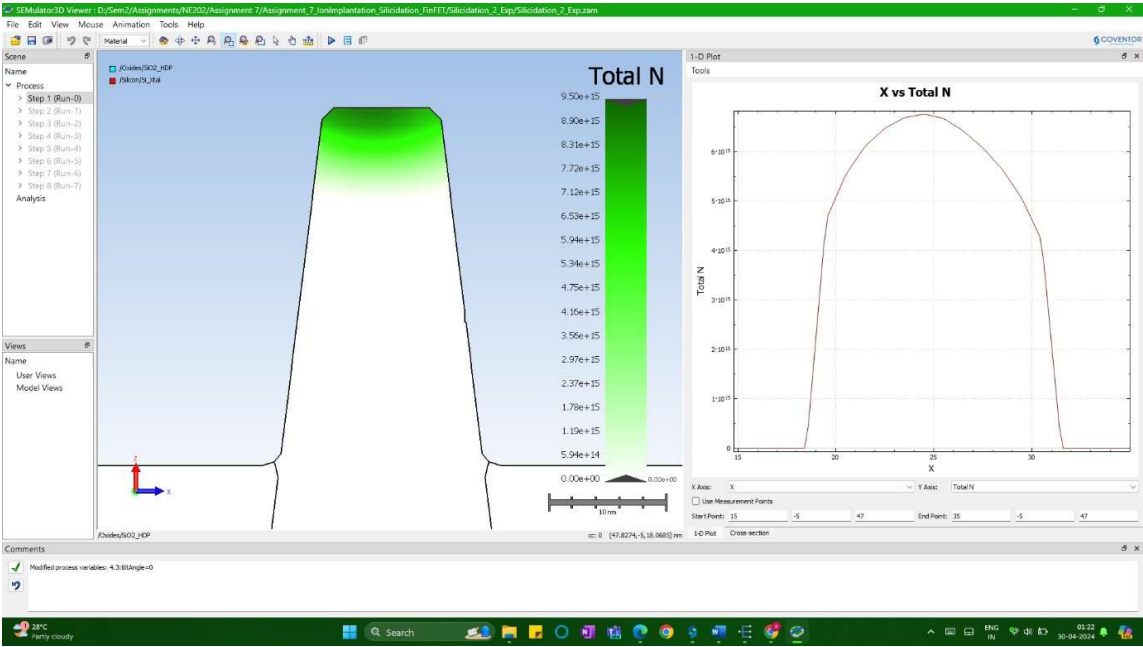


Figure 12 : Symmetric case (tilt angle = 0°)

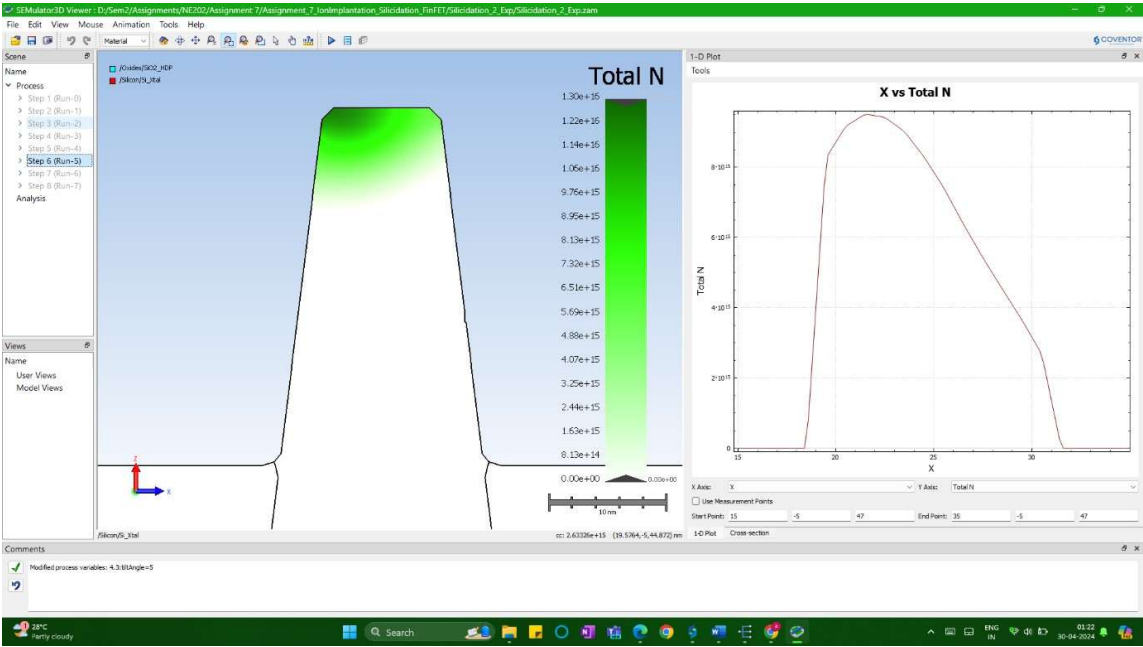


Figure 13 : Asymmetric case (tilt angle = 5°)

Question 2: Silicidation and other processes

Silicidation is the process to create a layer of silicide i.e. silicon mixed with a metal on desired area where metal contacts must be formed. This introduces a compound material with much lower contact resistance hence reduced current crowding than a metal layer directly deposited to create a contact.

Figure 14 shows broader process for silicidation.

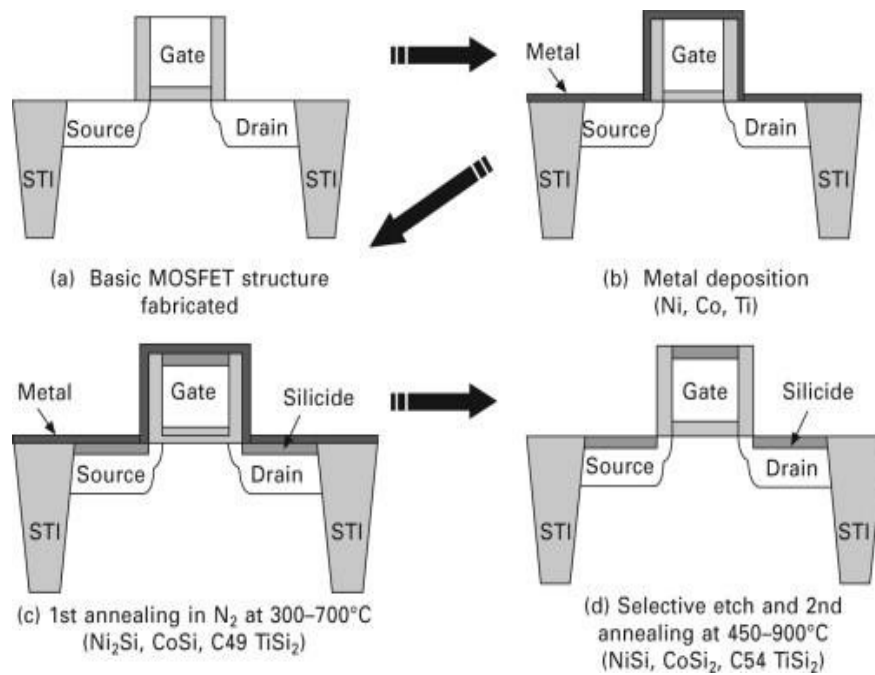


Figure 14 :Silicidation Process

Despite some drawbacks, such as intricate manufacturing processes and limitations due to high temperatures, silicide remains crucial for shrinking device sizes and improving performance in modern electronics. This is because it significantly reduces resistance and offers additional advantages like thermal stability.

Question2(a)

Salicidation:

Salicide is complex a self-aligned technique for forming conductive metal silicide on key transistor regions during fabrication. It deposits a metal layer that reacts with exposed silicon in the gate and source/drain areas under high temperatures, eliminating complex patterning and unreacted metal is then removed.

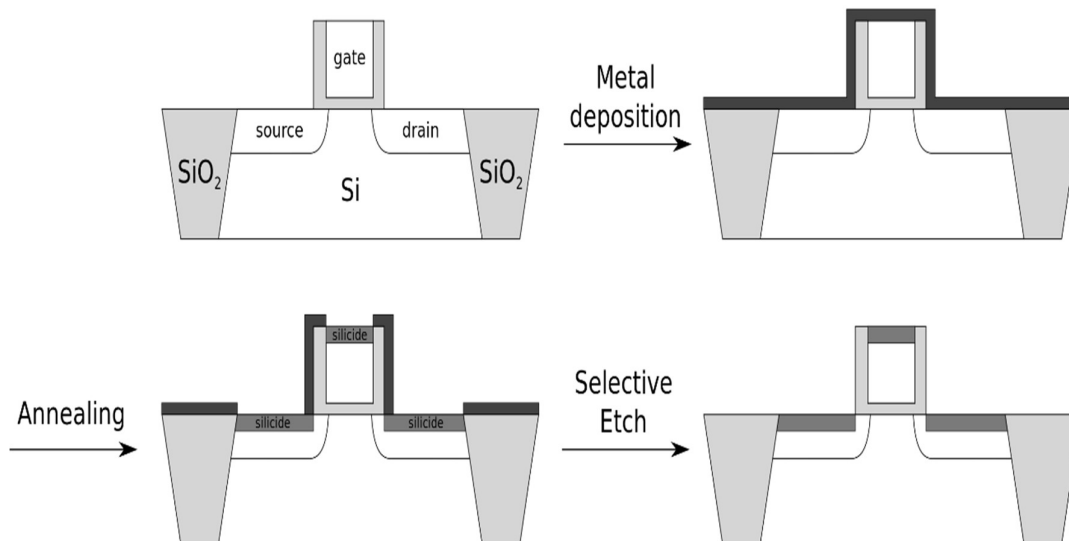


Figure 15 : Salicidation Process

Salicide is preferred for small transistors due to lower resistance but requires careful thermal management and layout planning for avoid bridging.

Polycide:

Polycide is a simpler process where the metal silicide is formed on the gate (polysilicon) before etching the transistor. This offers good thermal stability but only reduces gate resistance. Here, a metal layer is deposited that reacts with the polysilicon layer over gate oxide and then device is etched into its final form.



Figure 16 : Formation of Polycide

Polycide is preferred for processes requiring high thermal stability and avoids bridging but offers higher overall resistance.

Question2(b)

Silicides offer several compelling advantages over simple metal-silicon contacts:

- **Lowers Contact Resistance:** The reaction between metal and silicon creates a compound with better conductivity hence resistance at the contact point.
- **Controls Current Crowding:** Silicides offer lower resistivity and can result in a smoother interface. This translates to a more efficient flow of current between the metal and silicon. With lower resistance, the current doesn't get concentrated at specific points within the contact area, leading to a more even distribution – reducing current crowding.
- **Increases thermal stability:** Silicides are generally more resistant to high temperatures. This allows devices to function reliably at higher operating temperatures or during fabrication processes involving heat.
- **Improves adhesion:** The silicide layer forms a stronger bond with the silicon substrate compared to metal layer. This reduces the risk of separation or shorting during processing.

Question2(c)

The choice of metal used for silicidation plays a significant role in the final properties of the silicide layer. Here's some commonly used metals:

Table 3 : Common silicides with anneal temperature

Sl.No.	Metal	Silicide formed	Annealing Temperature (°C)
1.	Nickel (Ni)	NiSi, NiSi ₂	400 – 800
2.	Cobalt (Co)	CoSi, CoSi ₂	500 – 800
3.	Titanium (Ti)	TiSi ₂	600 – 800
4.	Tungsten (W)	WSi ₂	600 – 900
5.	Molybdenum (Mo)	MoSi ₂	600 – 1000

The selection of the appropriate metal for silicidation depends on various factors, including the desired resistivity, the thermal budget limitations of the overall process, and compatibility with other materials used. Sometimes, if required multiple metal layers are also used.

Question 3: 14nm advanced logic process

This task is centred around 14 nm advanced logic process, we aim to pinpoint the specific steps where silicidation occurs. After that we try to identify the material used for reducing contact resistance within the FET.

Figure 17 shows process flow for this simulation.

Number	Layer	Action	Name	Materials	Dependent	Depth	Mask	Mask Priority	Wafer	Wafer Size	Lot/Lot Size	Step Properties
1	Pre-Deposition	Pre-Deposition	Pre-Deposition	Pre-Deposition	Pre-Deposition	Pre-Deposition	Pre-Deposition	Pre-Deposition	Pre-Deposition	Pre-Deposition	Pre-Deposition	Pre-Deposition
2	Deposition	Deposition	Deposition	Deposition	Deposition	Deposition	Deposition	Deposition	Deposition	Deposition	Deposition	Deposition
3	Etch	Etch	Etch	Etch	Etch	Etch	Etch	Etch	Etch	Etch	Etch	Etch
4	Growth	Growth	Growth	Growth	Growth	Growth	Growth	Growth	Growth	Growth	Growth	Growth
5	Lithography	Lithography	Lithography	Lithography	Lithography	Lithography	Lithography	Lithography	Lithography	Lithography	Lithography	Lithography
6	Measurement	Measurement	Measurement	Measurement	Measurement	Measurement	Measurement	Measurement	Measurement	Measurement	Measurement	Measurement
7	FET Resistor	FET Resistor	FET Resistor	FET Resistor	FET Resistor	FET Resistor	FET Resistor	FET Resistor	FET Resistor	FET Resistor	FET Resistor	FET Resistor
8	Transistor/Contact	Transistor/Contact	Transistor/Contact	Transistor/Contact	Transistor/Contact	Transistor/Contact	Transistor/Contact	Transistor/Contact	Transistor/Contact	Transistor/Contact	Transistor/Contact	Transistor/Contact
9	Post-Process	Post-Process	Post-Process	Post-Process	Post-Process	Post-Process	Post-Process	Post-Process	Post-Process	Post-Process	Post-Process	Post-Process

Figure 17 : Steps involved in FinFET simulation

Resultant structure is shown by Figure 18 and Figure 19.

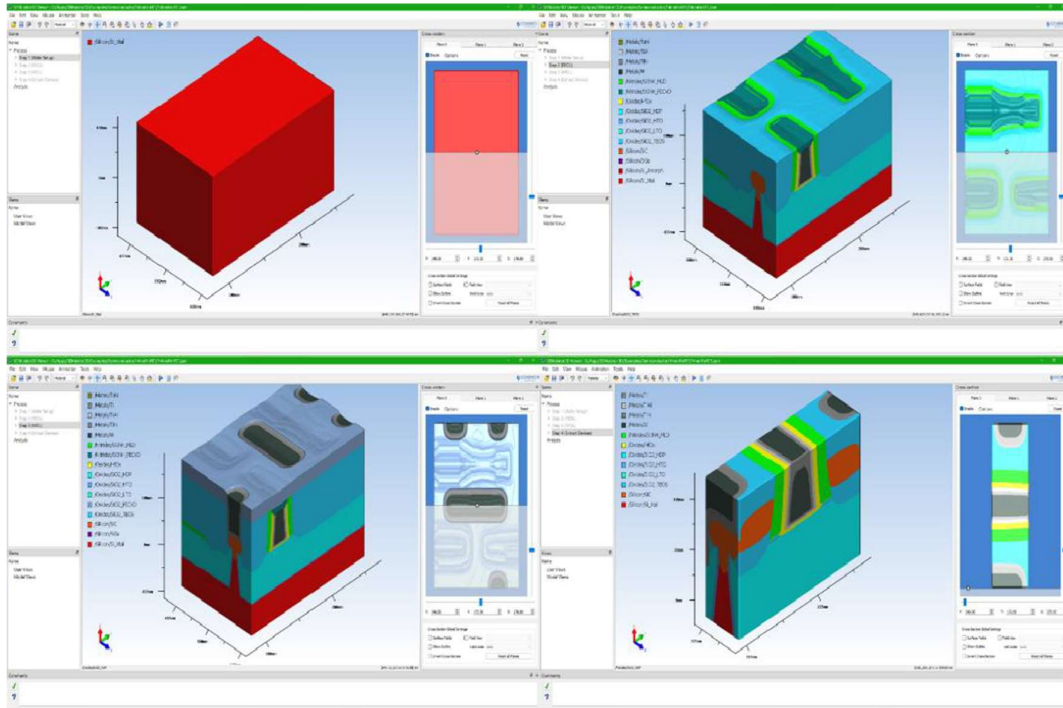


Figure 18 : Simulated 14nm FinFET process (3D view)

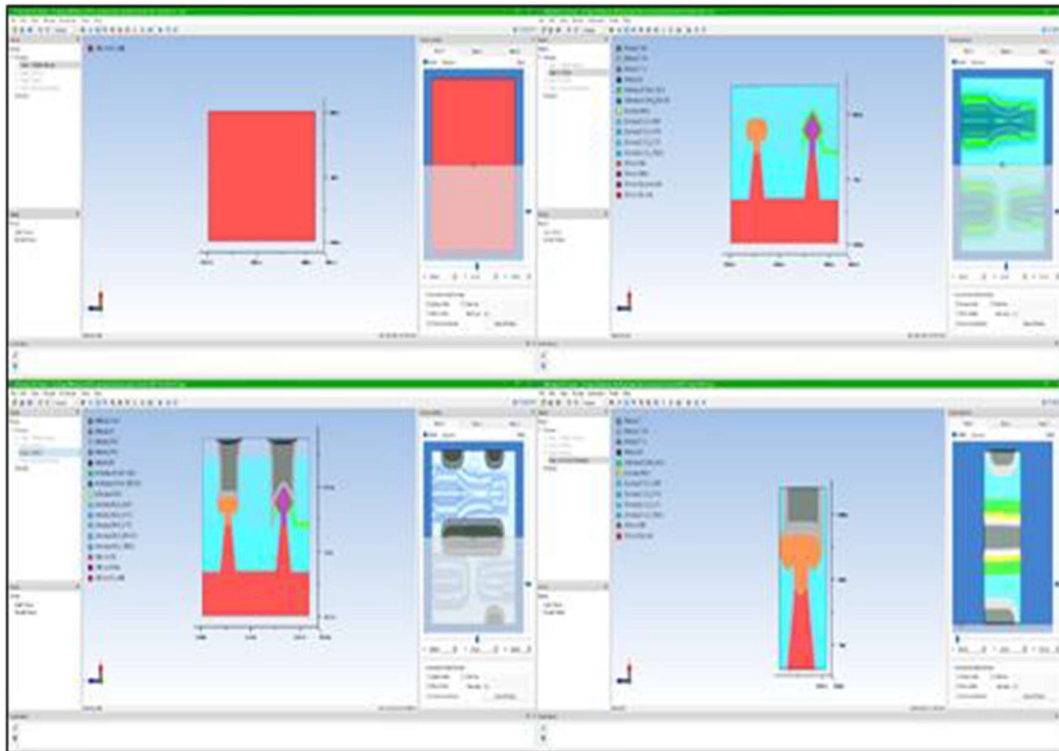


Figure 19 : Simulated 14nm FinFET process (Front View)

Question3(a)

In the provided process flow, step 3.9 deposits the metal (Ti) (Figure 20) needed for silicidation but it lacks a crucial high-temperature annealing step. Annealing step after deposition is essential for the Ti to react with the underlying silicon and form the desired silicide layer. Without this step, the titanium layer might not function as intended.

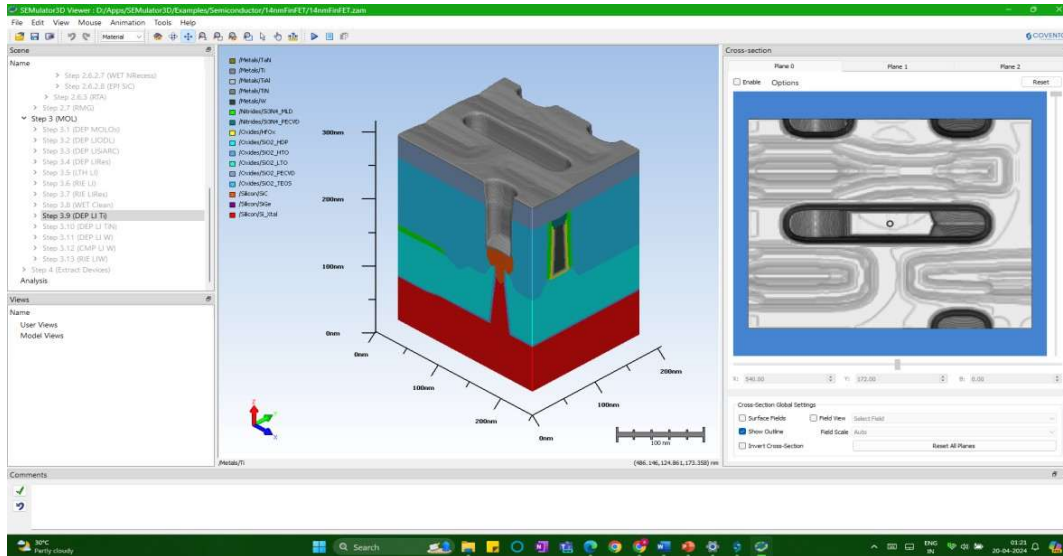


Figure 20 : Deposition of Ti

Question3(b)

The materials used for reducing source and drain resistance are SiC for NMOS in step 2.6.2.8 and SiGe for PMOS in step 2.6.1.9, shown in Figure 21. These materials are grown epitaxially and will not form silicide on further annealing.

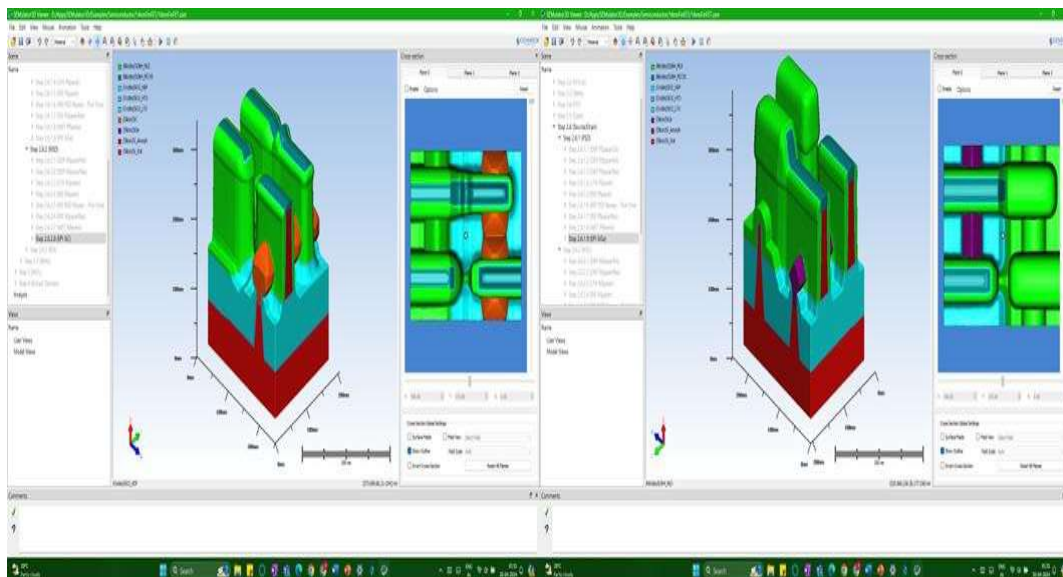


Figure 21 : Materials deposited for reducing resistance

Conclusion

This report investigated the critical role of wafer tilt during ion implantation for achieving proper doping profiles in FinFETs. Additionally, it highlighted the advantages of silicidation in reducing contact resistance and current crowding hence improving device reliability. By understanding these factors, we can optimize the FinFET fabrication process for better performance and scalability in future generations of integrated circuits..

References

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